

Collection of problems for Lectures 6-9

Below are a collection of 57 problems related to the material Lectures 6 through 9. At the end of Lectures 6, 7 and 8, I will assign a few questions from this set as the group assignment. Answers to the each assignment are due before the start of the next class. We will post solutions for all of these problems as the week progresses. There will be no assignment for the material in Lecture 9, but please study the questions and suggested solutions when preparing for the final exam.

These are group assignments. Each group should submit one solution set for each assignment. The name of every group member who participated in the exercise should be printed at the top of the assignment. One person from the group must attest that the names of the participating members are accurate. If possible, that person should sign the document. You may hand write (rather than type) your solutions – whichever is more convenient. Then scan/photograph your solutions and upload them to the canvas course website.

Problem 1 Consider the taxicab example from class. Suppose that the market demand increases to

$$P(Q) = 1.2 - \frac{Q}{100,000},$$

but everything else remains unchanged. That is, there are 6 non-transferable taxi licenses and each taxi driver has the same opportunity cost function given by $C(q) = 0.20q$ with a capacity of 10,000 miles. What is (a) the equilibrium price, (b) the economic profit each firm earns from owning the license? (c) the deadweight loss of restricting the number of licenses to 6?

Problem 2 In the competitive market for DVDs, market demand is given by $Q^d = 500 - 20p$ and quantity supplied is given by $Q^s = 5p$, where quantity is given in millions of DVDs sold per year.

- (a). What is the market equilibrium price and quantity?
- (b). Suppose the government decided to tax DVD sellers by an amount of \$5 per DVD sold.
 - (1) Under the tax, what after-tax price do consumers pay? (2). What after-tax price do sellers receive? (3). What is the new equilibrium quantity?
- (c). Describe, in words or with graphs, how does this tax generate a deadweight loss? That is, what is the source or story behind the deadweight loss?

Problem 3 True, False, Uncertain? [Explain your answer.] In a paper by University of Chicago Professor Sam Peltzman, it is shown that for supermarkets, approximately 40% of

any reduction in marginal costs is passed along to consumers through a reduction in prices. *Therefore, supply is relatively more elastic than demand.*

Problem 4 True, False, Uncertain? [Explain your answer.] The State of Illinois removes a 5 cent per gallon tax on gasoline (legally paid by the gasoline suppliers) because gasoline prices reach record highs. The demand elasticity of gasoline is -0.5 . The supply elasticity of gasoline is 1.5 . *The majority of the tax reduction to gasoline suppliers will be passed onto consumers.*

Problem 5 Assume that the short-run market demand curve for cartons of cigarettes is given by the equation:

$$Q^d = 100,000,$$

while the supply curve of cartons of cigarettes is given by the equation:

$$Q^s = 20,000 + 20,000p.$$

Assume that the government imposes a tax of \$1 per carton on the purchase of cigarettes.

- (a). What is the equilibrium price and quantity before the tax?
- (b). What is the total transaction price, p_s , received by the sellers, what is the total transaction price, p_b , paid by the buyers, and what is the after-tax quantity transacted?
- (c). What is the excess burden (deadweight loss) of this tax? Explain economically.
- (d). Assume that the demand of cigarettes is more elastic in the long run than in the short run. (i) How does the size of excess burden change from the short run to the long run? (ii) How does the distribution of the tax burden change from the short run to the long run? Briefly explain each.

Problem 6 Assume that the market demand curve for honeydew melons is given by the equation:

$$Q^d = 100,000 - 40,000p,$$

while the short-run supply curve of these melons is given by the equation:

$$Q^s = 20,000.$$

Assume that the government imposes a tax of \$1 per melon on the purchase of melons.

- (a). What are the equilibrium price and quantity before the tax is imposed?
- (b). What are the elasticities of supply and demand at the pre-tax equilibrium price and quantity from (a) above?
- (c). What are the equilibrium price and quantity after the tax is imposed?

- (d). How much tax revenue is raised by the government?
- (e). How much of the economic incidence of the tax is borne by consumers and how much by producers?
- (f). What is the deadweight loss of this tax? Explain.
- (g). Assume that the supply of melons is more elastic in the long run than in the short run. How does the size of the deadweight loss change from the short run to the long run? How does the distribution of the economic tax incidence change from the short run to the long run? Explain.

Problem 7 True, False, or Uncertain. [Explain your answer!] It has recently been argued that consumers see very little of gasoline taxes passed along into the price. Assume that the gasoline market is perfectly competitive and the taxes are levied per gallon. *This result implies that the supply of gasoline must be very elastic.*

Problem 8 True, False, or Uncertain. [Explain your answer!] If the price of steel sold by the steel mills is held below the market equilibrium by a government regulation (but the resale of steel is unregulated by the government), *then price of automobiles will rise.*

Problem 9 Suppose that the demand and supply for coffee are given by equations

$$Q^d = 1200 - 200p$$

$$Q^s = 200p.$$

- (a). Find the equilibrium price and quantity for this market.
- (b). Suppose that the government decides to buy up as much coffee as it takes to raise the market price to $p = 4$. Furthermore, suppose the government destroys all of its purchased coffee. How much deadweight loss does this intervention generate?
- (c). Suppose instead that the government decides to subsidize coffee production by paying producers \$1/bushel for every bushel of coffee sold. What are the after-subsidy equilibrium prices that the buyers effectively pay, p_b^* , and that the sellers effectively receive, p_s^* ?

Problem 10 True, False, or Uncertain. [Explain your answer!] Consider a constant-cost industry with homogeneous firms in long-run equilibrium, with price equal to $p = 1$. Suppose that the government places a price ceiling on output at $p = \frac{1}{2}$, and that this law is effective at preventing trade from occurring at higher prices than $p = \frac{1}{2}$. *The long-run equilibrium output of this market is unchanged by the price restriction.*

Problem 11 True, False, Uncertain? [Explain your answer.] Suppose that demand for olives is elastic and the government decides to encourage production by offering olive growers

a subsidy for each kilo of olives sold at market. *Total market expenditures by consumers on olives decrease following the subsidy.*

Problem 12 Consider a perfectly competitive labor market with downward sloping labor demand (from firms) and upward sloping labor supply (from workers). Suppose that the government institutes a minimum wage law which requires firms to pay $\underline{w}$ per hour which is more than the current market equilibrium of w^* per hour. Indicate in a graph (or graphs) how the market outcome changes (e.g., wages, labor transacted, etc.) and indicate how surpluses change for all market participants (workers and firms, in and out of the market). Also illustrate the deadweight loss from the wage law and indicate (verbally) why it emerges?

Problem 13 Which of the following statements is *NOT* generally true regarding a binding price ceiling in a competitive market? Explain your answer. [No credit will be given without a suitable explanation.]

- (a). All producer surplus is transferred to consumers.
- (b). The market will not clear due to excess demand of the good.
- (c). Producer surplus decreases when compared to the market before the price ceiling.
- (d). Consumer surplus may increase or decrease.

Problem 14 True, False, Uncertain? [Explain your answer.] *If a government wishes to minimize the deadweight loss of raising a fixed amount of tax revenues, it should focus its taxes on goods whose demand or supply are relatively inelastic.*

Problem 15 True, False, or Uncertain. [Explain your answer!] *In a perfectly competitive, constant-cost industry, the imposition of a tax of \$2/unit will cause the buyer's total transaction price of the good (price paid to seller plus tax paid to government) to rise by \$2 in the long run.*

Problem 16 Suppose that a monopolist's *opportunity* costs are given by the function $C(q) = 16 + 4q$ and the monopolist's demand is given by $q^d = 40 - 2p$. What is the firm's optimal choice of p and q ?; what are the corresponding profits?

Problem 17 True, False or Uncertain? [Explain your answer.] A monopolist with constant marginal cost currently sells to a market with demand function given by $q^d = D(p)$. *If the monopolist's market size doubles so that demand becomes $2D(p)$ (i.e., twice as much is demanded as before for any given price), then monopolist's optimal price will also double.* Hint: Consider the case of linear demand, $D(p) = a - bp$, to get intuition.

Problem 18 True, False, or Uncertain? [Explain your answer!] *In the long run, a monopolist will choose to produce at the point where long-run average cost is minimized.*

Problem 19 The local monopoly coffee and espresso shop sells regular coffee (denoted with a “c”) and premium espresso (denoted with an “e”). The market demand functions for coffee and espresso are, respectively,

$$q_c^d = 12 - 3p_c + p_e,$$

$$q_e^d = 12 + p_c - 3p_e,$$

where quantities are measured in cups per week and prices are measured in dollars per cup. Note, we can also write this demand system as a pair of demand “curves” with prices on the lefthand side:

$$p_c = 6 - \frac{3}{8}q_c - \frac{1}{8}q_e,$$

$$p_e = 6 - \frac{1}{8}q_c - \frac{3}{8}q_e.$$

(a). Are these products substitutes or complements? Explain.

(b). Suppose that the marginal and average costs of producing both products are constant. Regular coffee costs \$0.50 per cup to produce, and espresso costs \$1.50 per cup to produce. What are the profit-maximizing prices of regular coffee and espresso? (Hint: Write the monopolist’s profit function as either a function of quantities (q_c and q_e) and maximize with respect to these, or write the monopolist’s profit as a function of prices (p_c and p_e) and maximize with respect to these. The second approach is more direct for this question where we are asked for the optimal price and the marginal costs are constant.)

Problem 20 A monopolist’s demand curve is given by $p = 1000 - q$. Production costs are $C(q) = 30,000 + 100q + 0.5q^2$. What is the monopolist’s revenue function? What is the monopolist’s profit function? What level of output maximizes the monopolist’s profits, assuming that the monopolist produces? Should the monopolist produce? What are the monopolist’s profits under the optimal decision?

Problem 21 If a monopolist maximized profit per unit, will total profits be maximized? Explain why or why not.

Problem 22 True, False, Uncertain? [Explain your answer.] *The ratio of a monopolist’s optimal price to its marginal cost is larger when the market elasticity of demand is greater in absolute value (i.e., more elastic).*

Problem 23 True, False, or Uncertain? [Explain your answer!] *Whenever a firm changes its price in one direction, its revenue changes in the opposite direction if the demand curve is relatively inelastic.*

Problem 24 True, False, or Uncertain? [Explain your answer!] *If the Maximum Pie Company (which is a price-making monopoly) wants to maximize its profit, it should follow this motto: Whenever you can sell an additional pie for more than it costs to make, make more. Whenever you must sell a pie for less than it cost to make it, make less.*

Problem 25 True, False, or Uncertain. [Explain your answer!] When an author sells exclusive rights to a book to a publisher, the publisher becomes a monopoly seller of the book. Assume that, in return for the exclusive rights, the publisher pays the author 10% of the gross revenues from the book (i.e., the author receives $\frac{1}{10}pq$). Assume that demand for the book is neither infinitely elastic nor completely inelastic and that the marginal cost of each additional book is greater than zero. *The author will feel that the publisher charges too much for the book.*

Problem 26 Consider a monopolist with demand given by the relationship $q = 20 - 2p$ and a cost function given by $C(q) = 5q$.

- (a). What is the optimal price and quantity for the monopolist?
- (b). Illustrate in one figure the MR, MC and demand curves. (You should label the axes and indicate the values of the curves wherever they intersect the axes.) Indicate the deadweight loss from monopoly.
- (c). If the monopolist could perfectly 1st-degree price discriminate, how does the monopolist's marginal revenue curve change? What is the new optimal quantity?

Problem 27 The local government decides that it wants cable television in the area. Suppose that the demand curve for cable TV is given by $P(q) = 500 - q$ and that the economic cost for operating a cable company is $C^O(q) = 10,000 + 100q$, where q is the number of households deciding to buy cable TV and p is the price per year for a cable hookup. [Ignore any differences between short-run and long-run costs.]

- (a). If the city cares only about increasing city revenues through licenses, should it sell a license to operate cable TV to one firm or several? To achieve this end, should it regulate the price as part of the terms of the license? [Assume that firms will know how many licenses will be sold and whether or not they'll be regulated when they make their decision.]
- (b). How much cash can the government raise from the sale of these cable TV rights? [Solve for the dollar amount for full credit.]
- (c). Make an economic argument (no need to use math) that if the city knows the cost function of the cable TV industry and if it cares about the sum of its consumers' surplus and the revenues it makes from licensing, that it will wish to license a single cable company with some sort of regulation (e.g. price ceilings) on the price which the cable TV company can charge its customers. In your explanation, indicate why licensing to one firm with price regulation is preferred to licensing to several firms.

Problem 28 Consider a monopolist with opportunity costs given by $C^O(q) = 2q + 2$ and demand given by $q = 20 - 2p$. What is the profit maximizing quantity and price, and corresponding profit?

Problem 29 True, False, or Uncertain. [Explain your answer!] *A monopolist that wishes to maximize profit will always find it optimal to set a price such that all consumers who value the product above marginal cost will purchase the product.*

Problem 30 True, False, or Uncertain? [Explain your answer!] An empirical study of demand elasticities across various industries finds that with monopolies, demand is elastic (i.e., $|\varepsilon| > 1$). *This empirical finding implies that monopolies can only profitably exist in markets in which demand is elastic at any price.*

Problem 31 Suppose that a monopolist's (opportunity) costs are given by the function $C(q) = 10 + 8q$ and the monopolist's demand is given by $q^d = 10 - \frac{1}{2}p$. What is the firm's optimal choice of p and q ?; what are the corresponding profits?

Problem 32 True, False, Uncertain? [Explain your answer.] *A profit maximizing monopolist's selling price is always lower than its marginal revenue.*

Problem 33 Suppose that a monopolist faces demand $P(q) = 300 - 6q$ and has marginal cost of $MC(q) = 120 + 6q$.

- (a). How many units should this monopolist produce to maximize profits?
- (b). What price should this monopolist charge to maximize profits?
- (c). What is the dead-weight loss of monopoly in this case?

Problem 34 Suppose that a dominant firm with marginal costs and average costs both equal to 10 faces a competitive fringe of price-taking firms that will supply inelastically exactly 50 units of output. The goods are homogenous, and the market demand function is $Q^d = 200 - 3p$. How much should the large firm supply?

Problem 35 Suppose that a monopolist sells a single good to a population of consumers which is comprised of two types which arise in equal proportion. One type of consumer (type A) values the good at 100; the other type of consumer, type B , values the good at only 75. Each consumer wants at most one unit of the good. Each type of consumer occurs with equal proportion the population. (If it helps, you can let n be the even number of consumers and $n/2$ be the number of each type in the market.)

- (a). Assume that the monopolist finds it profitable to operate. If the monopolist's marginal cost of production is 0 and he cannot distinguish between the consumer types, what is the optimal price for the monopolist to charge? (You may assume that a consumer who is indifferent between purchasing and not purchasing will always choose to purchase the good.)
- (b). Suppose that the monopolist has a choice between three product designs, all with identical setup costs. Once the product design is chosen, the monopolist sells only that

variation of the good. (The monopolist is not designing a product line in this question.) Marginal costs of production are zero under each design. The first product design is the same as before (i.e., type A consumers value the good at \$100 and type B consumers value it at \$75). The second product design appeals to type A consumers: type- A consumers value the good at \$125 while type- B consumers value this design the same as the first product design, at \$75. The third product design appeals more to type- B consumers: type- B consumers value this variant of the good at \$80. Type A consumers, unfortunately, like the third design least of all: they value this version of the good at only \$85. If all these product designs are equally costly to implement and only one can be chosen, and if the marginal costs of production of the good is zero regardless of the design, which product design should the monopolist choose?

(c). Continuing with the facts of part (b): From the point of view of society, which product design generates the most welfare (consumer surplus plus economic profit)? [Hint: the answers to (b) and (c) are different.] Explain this conclusion in contrast to your answer to (b).

Problem 36 Suppose that a dominant firm with marginal costs of $MC(q_m) = 50$ faces a competitive fringe of price-taking firms will to supply according to the supply function $S_f(p) = 150p$. The goods are homogenous, and the market demand curve is

$$p = 600 - \frac{1}{50}Q.$$

What is the monopolist's optimal output?

Problem 37 Suppose that there are two duopolists, “A” and “B”, choosing output in a Cournot game. The market demand curve is given by

$$p = 100 - q_a - q_b.$$

Firm A has zero marginal cost, while firm B has a constant marginal cost of $MC_b = 20$. What is the equilibrium price in the market?

Problem 38 Suppose that there are two duopolists, “A” and “B”, choosing output in a Cournot game. The market demand curve is given by

$$p = 100 - q_a - q_b.$$

Firm A has zero marginal cost up to a capacity of 15, beyond which it cannot produce. Firm B has a constant marginal cost of $MC_b = 20$ and can produce unlimited amounts at that unit cost. What is the equilibrium price in the market?

Problem 39 Kate and Alice are ready-mix concrete duopolists. The market demand function is

$$Q = 20,000 - 200p,$$

where p is the price of a cubic yard of concrete and Q is the number of cubic yards demanded per year. The marginal cost of producing a cubic yard for each firm is constant at \$80. Kate and Alice compete by choosing quantities and the market price is determined so as to equate their supplies with market demand.

- (a) What is Kate's residual demand curve, as a function of Alice's expected supply, $\hat{q}_A$?
- (b) What is Kate's optimal quantity as a function of $\hat{q}_A$?
- (c) What is the equilibrium market supply of output, $Q^* = q_A^* + q_K^*$?
- (d) Suppose now that Kate can publicly commit to a particular output before Alice has a chance to choose her output. What is the (Nash) equilibrium market price in this Leader-Follower quantity game?

Problem 40 Consider a monopolist with demand function $q = 150 - 3p$, and opportunity-cost function, $C^O(q) = 10q$.

- (a). What point on the demand curve should the monopolist choose (i.e., what p^* and q^*).
- (b). What is the deadweight loss of monopoly in this case?

Problem 41 Consider a dominant firm competing against a competitive fringe of price-taking firms. The supply from the competitive fringe is given by

$$S_f(p) = 19p.$$

The market demand curve is given by

$$p = 1200 - Q,$$

where $Q = q + S_f(p)$. If the dominant firm has constant marginal cost equal to 20, how much should it produce?

Problem 42 Consider two competing duopolists, Tina and Emily. The both sell a homogeneous good into a market with demand given by

$$p = 200 - \frac{1}{2}Q.$$

Emily's marginal cost is constant and equal to $MC_e = 15$. Tina has a lower MC cost which is constant at $MC_t = 10$. Tina and Emily play an output game in which they both bring their output to market and the market price is determined from above with $Q = q_e + q_t$ (i.e., this is a Cournot output game).

What is the equilibrium price?

Problem 43 Price discrimination requires the ability to sort customers and the ability to prevent arbitrage. Explain how the following can function as price discrimination schemes and discuss both sorting and arbitrage.

- (a). requiring airline travelers spend at least one Saturday night away from home to qualify for a low fare.
- (b). selling food processors along with coupons that can be sent to the manufacturer to obtain a \$10 rebate.
- (c). offering temporary price cuts on bathroom tissue.
- (d). charging high-income patients more than low-income patients for plastic surgery.

Problem 44 (Overheard at a grocery store). “I am getting sick and tired of all these specials, double coupons and the like. If the store would simply discontinue these specials and pass the savings along in lower prices every day, I would be better off.” Explain whether you agree or disagree with this statement.

Problem 45 You are an executive for Super Computer, Inc. (SC), which rents out super computers. SC receives a fixed rental payment per time period in exchange for the right to unlimited computing at a rate of p cents per second. SC has two types of potential customers of equal number – ten businesses and ten academic institutions. Business customers have demand functions $q_B = 10 - p$, where q_B is in millions of seconds per month; academic institutions have demands $q_A = 8 - p$. The marginal cost to SC of additional computing is two cents per second, no matter what the volume. Ignore shutdown decisions.

- (a). Suppose that you could separate business and academic customers. What rental fee and usage fee would you charge each group? What are your profits?
- (b). Suppose you were unable to keep the two types of customers separate, and you charged a zero rental fee. What usage fee (i.e., price) maximizes your profits? What are your profits?
- (c). Suppose you set up one two-part tariff; that is, you set one rental and one usage fee that both business and academic customers face. What usage and rental fees will you set? What are your profits? Explain why price is not equal to marginal cost.

Problem 46 Suppose that a firm has two distinguishable customers: students and non-students. The total demand for students is $q_s = 100 - p$. The total demand for non-students is $q_n = 200 - p$. Total costs are $C(q_s + q_n) = \frac{1}{2}(q_s + q_n)^2$. What is the optimal pair of prices which a 3rd degree price-discriminating monopolist would choose? What are profits?

Problem 47 True, False, Uncertain? [Explain your answer.] A Belgian ice cream company decides to expand its sales to Luxembourg. Because the company is less known outside of Belgium, it decides to charge a lower price for its Luxembourg ice cream than for its Belgium ice cream. Assume that it is not profitable for an outside company or individual to arbitrage ice cream across national boundaries. *The price the company charges for its Belgian ice cream should be derived independently of its price for its Luxembourg ice cream.*

Problem 48 Macroware has two types of customers for its computer design software: home users and business users. Each user will purchase only one unit of the software.

Macroware is considering producing two versions of the software. Version A contains the full features of the Macroware software. Version B is the same as Version A in every way, except it does not allow the user to design in color. The willingness to pay of business and home users for Macrowares software is given below:

Version	Business	Home
Version A	\$2000	\$800
Version B	\$1000	\$650

There are 100 business users and 100 home users.

For simplicity, assume also that Macroware incurs zero costs for either version.

(a). If Macroware can tell Business and Home users apart and can charge them different prices, what price will Macroware set for Version A for home users? What price will Macroware set for Version A for business users? Will it also sell version B? If so, at what prices; if not, then explain why not?

(b). Now suppose that Macroware cannot tell business and home users apart and cannot charge them different prices for the same good. However, Macroware can charge different prices for Version A and Version B of the software. How much will Macroware charge for Version A? How much will Macroware charge for Version B?

(c). How might your analysis change in part (b) change if there were 190 business users and 10 home users? (Do not do any calculations, just describe).

Problem 49 Third-degree (or direct) price discrimination tends to be more common in the sale of services (movies, transportation, medical services, etc.) rather than manufactured goods (TV's, stereos, food, etc.). What is a *simple* economic explanation for this?

Problem 50 Suppose that there are two consumers in equal proportion in the marketplace: one which is type A and one which is type B. You are a monopolist, your costs of production are zero, and you cannot distinguish between the two consumer types. The consumer demands are characterized by the following data which lists the maximum prices the various consumers will pay for each additional unit:

Quantity	Consumer A	Consumer B
first	10	12
second	8	10
third	3	7
fourth	2	5

(a). If only a single per-unit price is offered, what is the profit-maximizing price? What is the total profit at this price? [Calculate profits assuming that there is exactly one consumer of each type.]

(b). If a different price can be offered for each level of output, what is the optimal price schedule to offer? (That is, what price will be charged for the first unit, for the second unit, etc.?) What is the total profit? [Calculate profits assuming that there is exactly one consumer of each type.]

Problem 51 Hayley's Bistro serves two types of customers, senior citizens and young people. There are 100 potential customers of each type. The total willingness to pay for dinner at Hayley's by the members of each group is a function of the time of day. Each customer will only dine once per day.

Time	Senior	Young People
Dinner at 6	\$29	\$45
Dinner at 7	\$35	\$58

Assume the costs of serving customers are constant throughout the day and are equal to \$10 per customer. This implies that overcrowding, turning people away, etc. are not problems, even at peak hours.

(a). Suppose that Hayley is forced to charge the same price for dinner at 6 as for dinner at 7. Hayley is also forced to charge the same price to Seniors as to Young People. How much should Hayley charge? How much profit will Hayley earn?

(b). Would your answer in (a) change if it cost \$20 to serve each customer?

(c). Now suppose that Hayley can charge a senior citizen price and a regular price. What are the profit-maximizing prices for senior citizens and young people? How many people total will dine at 6? At 7? What will Hayley's profits be?

(d). Now suppose that Hayley cannot charge different prices to seniors and young people. However, Hayley can charge different prices for dinner at 6 and for dinner at 7. How much should Hayley charge for dinner at 6? For dinner at 7? How much profit will Hayley earn?

(e). Hayley does some more market research for eating at 5pm. She learns that her diners have the following willingness to pay for dinner at various times. (Each diner still only eats at most once per night)

Time	Senior	Young People
Dinner at 5	\$17	\$18
Dinner at 6	\$29	\$45
Dinner at 7	\$35	\$58

Assume that Hayley cannot charge seniors and young people different prices, as in part (d). Also, she thinks that it would be too confusing to have many different prices throughout the evening. Thus, she will choose to offer only the 7pm (regular) time and an early time, each with different prices. Hayley must decide both the prices and what "early" means (5pm or 6pm).

What prices will Hayley charge and which dining time will be required for the lower "early" price? How much profits will Hayley earn?

Problem 52 The producers of consumer goods often issue coupons that let one buy the product at a discount. These coupons are usually distributed as part of an advertisement for the product; they may appear in a newspaper or magazine, or as part of a promotional mailing. For example, a coupon for a particular breakfast cereal might be worth 25 cents toward the purchase of a box of the cereal. We might wonder why firms issue these coupons when printing them and collecting them are expensive. We might wonder why the firm does not just lower the price of the item.

One possibility is price discrimination. Consumers who clip coupons may be more price sensitive than other people. By issuing coupons, a cereal company can separate its customers into two groups, and in effect charge the more price-sensitive customers a lower price than the other customers.

The price elasticities of demand for users vs. nonusers of coupons have been estimated (Narasimhan, "A Price Discrimination Theory of Coupons," *Marketing Science*, 1984). Here are some examples of the price elasticities of demand for users vs. non-users of coupons:

PRICE ELASTICITY	Nonusers	Users
Spaghetti sauces	-1.50	-2.00

Obviously, the elasticity of demand is not constant along the demand curve, but suppose that the elasticity is constant for the purposes of this question. [I've also taken the liberty of rounding Narasimhan's numbers for spaghetti sauce to make computations easy.]

Suppose that a monopoly manufacturer of spaghetti sauce has decided to charge \$2 for spaghetti sauce. What face value of coupon should the manufacturer issue? [Assume you are not concerned about non-users of coupons switching to become users of coupons.]

Problem 53 True, False, or Uncertain. [Explain your answer!] Consider a monopolist who price discriminates using third-degree (direct) price discrimination across two markets, *A* and *B*. If at the optimal prices the elasticity of demand is -2 in market *A* and -4 in market *B*, then the optimal prices will be in a ratio of 3 : 2.

Problem 54 Suppose that you are a monopolist with zero economic costs. You have two types of customers. Some consumers are more price sensitive than others (type *A*'s) and

these more elastic consumers also have more time available for collecting coupons for price reductions. The inelastic consumers (type B 's) can also collect coupons, but at a higher personal cost. Specifically,

	Consumer A	Consumer B
value of good	10	20
cost of coupon collection	2	5

For example, consumer B will pay 20 for the good without producing a coupon and 15 (i.e., $20 - 5 = 15$) for the good if consumer B has to go through the trouble of collecting the coupon.

(a). Suppose that your customer population consists of equal proportions of each type. What is the optimal price for the good? Do you offer a coupon? If so, how much of a price discount do you offer for the coupon?

(b). Suppose that your customer population consists of twice as many type B consumers as type A consumers. What is the optimal price for the good? Do you offer a coupon? If so, how much of a price discount do you offer for the coupon?

(c). Suppose that the cost of coupon collection is the same for both consumer types. Without doing any calculations, can you say whether or not coupons will be used to maximize profits? Explain.

Problem 55 Mitchell's movie theater has two types of customers: *students* who are not willing to pay much to see a movie at the theater, and regular customers ("*regulars*") who have a higher willingness to pay. There are 100 customers of each type. Mitchell has excess seat capacity (more than 200 seats) and can show movies at both 5pm and 8pm at no additional cost. Both types of customers prefer to watch movies at 8pm rather than 5pm. The willingness to pay for each type of customer at the two possible times is shown below.

	regulars	students
8pm	\$15	\$8
5pm	\$7	\$5

The marginal cost of an additional theater-goer is zero at either showing. Assume each individual attends at most one movie.

(a). Suppose Mitchell can tell students and regulars apart. Also suppose that he can set prices based on customer type (student or regular) and time of showing (5pm or 8pm). What prices will he set for each movie type and customer type? Explain. (Recall that a consumer will buy only one movie.) If he decides not to offer a particular time to a particular type of customer, indicate that as well. Explain.

(b). Now suppose that Mitchell cannot tell students and regulars apart. However, Mitchell can charge different prices for the 5pm and 8pm showings. How much will Mitchell charge for 8pm showing? How much will he charge for 5pm showing? Explain.

(c). Suppose that instead of 100 of each type of consumer, there are 150 student customers and only 50 regular customers. How does your answer to (b) change? Explain.

Problem 56 Suppose that a monopolist sells to two markets and can set a different price in each market. The demand functions are

$$q_1 = 200 - 2p_1,$$

$$q_2 = 300 - 2p_2.$$

The monopolist's marginal and average cost is zero up to a capacity of 200 units; the monopolist cannot produce beyond 200 units.

What are the optimal prices to set in each market?